

5.1 CLIP

Two towers and a cosine. CLIP scores pairs; it does not decode a sentence.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

CLIP (Radford et al., 2021) is the coordinated, contrastive model that made zero-shot vision–language the default. Stanford CS231N 2025 L16’s punchline is the one we keep: after training you have a **similarity score** between an image and a text, not a captioner.

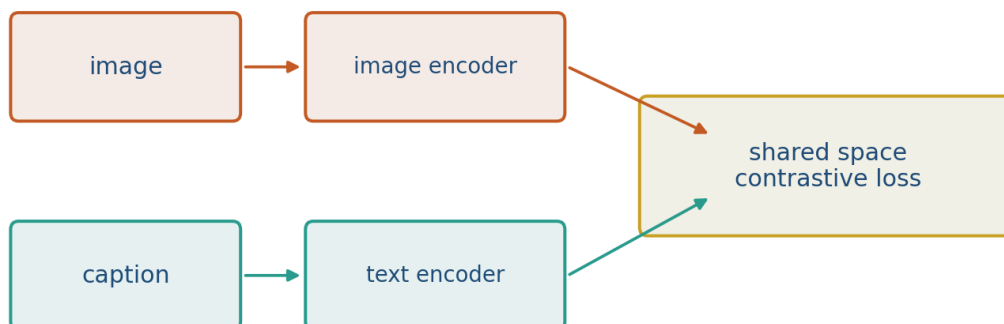
First time this method appears. **CLIP** is two towers and a cosine. It scores image–text pairs. It does not write a caption.

What. Image encoder + text encoder, InfoNCE on the batch, keep both towers. **Why.** One model that retrieves and zero-shot classifies without a detector or a decoder. **Architecture.** `encode_image`, `encode_text`, cosine. No `generate`. **How.** Train on web pairs. At test, embed the image and a list of prompts. Note 5.2 adds a decoder (BLIP). **Formula.** Same InfoNCE as 4.3 with learned τ . API is three calls. **Tradeoffs.** + Zero-shot and retrieval. – Not a captioner; hungry for batch size; prompt wording is method; bias next note.

1. Two towers

An **image encoder** (ResNet or ViT) and a **text encoder** (a transformer). Both emit a vector in the same dimension. Training: InfoNCE on the batch (note **4.3**). After training you keep both towers; you do not keep a fused head.

CLIP: two towers, one cosine, a batch of matches



The text tower never sees pixels. Retrieval still works because training put matched pairs at high cosine. A query string is just another text vector.

Open-source copies (OpenCLIP, SigLIP) change the loss or the data. The picture is stable. SigLIP replaces the softmax InfoNCE with a sigmoid on each pair; you still have two towers and a score.

Write the API on the board: `encode_image`, `encode_text`, `cosine`. Three calls. No `generate`.

2. What CLIP is good at, and what it is not

- **Zero-shot** classification via prompts (a photo of a dog).
- **Cross-modal retrieval**: image $\rightarrow$ text and text $\rightarrow$ image (note 5.3).
- A **frozen** image backbone for later captioners and diffusion models (Week 12).

What it is not: a captioner (it does not decode a sentence), a detector with boxes, or a guarantee of factual correctness. It scores pairs.

CS231N's next slide after CLIP is **CoCa**: add a decoder with a captioning loss. That generation job is note 5.2 (BLIP). Do not skip from CLIP to LLaVA in this hour.

Prompt wording is part of the method. CS231N: a single phrase can be too peaked; ensembling "a photo of", "a drawing of", ... is optional engineering, not a new model.

3. Temperature and batch size

Temperature τ (often learned, around 0.07) sharpens the batch softmax. Small τ makes off-diagonal competition brutal; that is why CLIP wanted huge batches.

CS231N's first listed disadvantage of CLIP-style models: they **rely on batch size** to learn fine-grained concepts. A batch of 8 has 7 negatives. A batch of 32 has 31 negatives. You still will not see both "a mug in grass" and "grass in a mug" in one batch; compositionality is a later paper, not this lecture.

Hugging Face `CLIPModel` plus a processor is enough for Lab 5's *idea*; the lab itself uses a toy space so nobody waits on a download. If a project cites "CLIP," name the Hugging Face id. Freezing is allowed. Pretending freeze is fine-tune is not.

4. Teaching this note

30–40 minutes. Two towers, cosine, "not a captioner," then the τ and batch-size arithmetic. Play the **CLIP paper video** in class as **selected chapters**: towers **4:40–9:00**, contrastive **14:40–22:25**. Students can finish the hour as homework. Lab 5 is a 2-D fake CLIP, not OpenAI weights.

Minute plan: 10 min two towers; 10 min worked $N = 2$ and τ ; 5 min "cannot decode a sentence"; 12 min video chapters. Assign robustness to note 5.3.

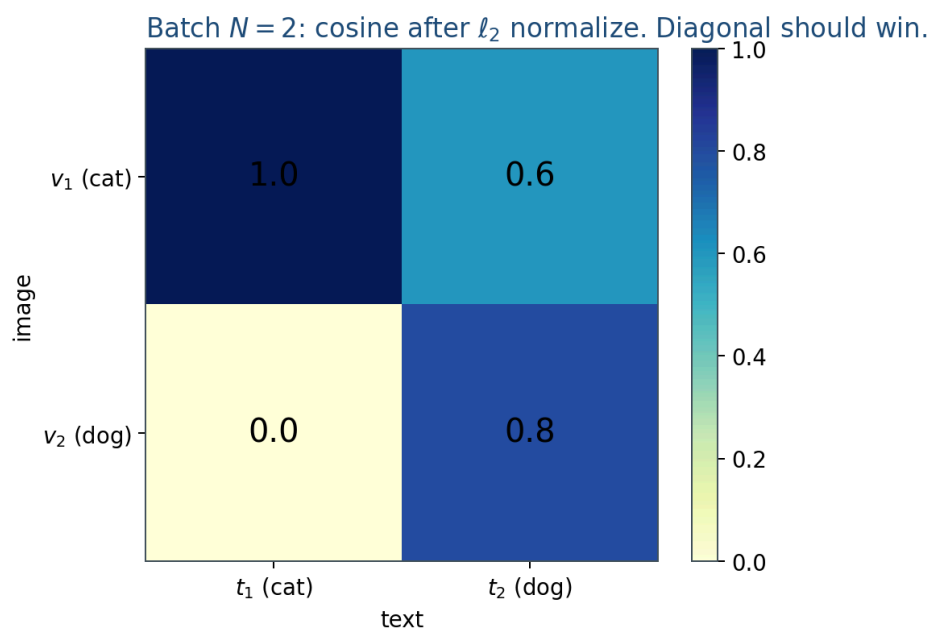
5. Worked example

Batch $N = 2$. Already ℓ_2 -normalized:

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, t_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, t_2 = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}.$$

Cosines (the matrix S):

$$S = \begin{bmatrix} 1.0 & 0.6 \\ 0.0 & 0.8 \end{bmatrix}.$$

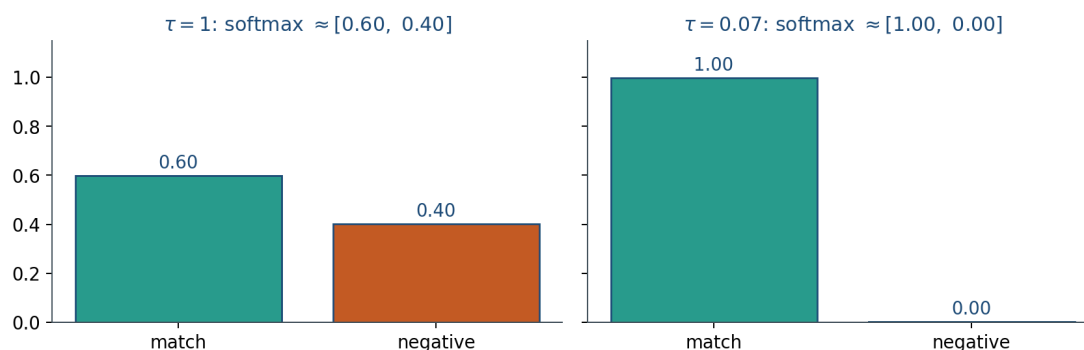


InfoNCE on **row 1** uses logits S_{1j} / τ . With $\tau = 1$,

$$\text{softmax}([1.0, 0.6]) \approx [0.60, 0.40], \quad -\log 0.60 \approx 0.51.$$

With $\tau = 0.07$, the logits are $[14.3, 8.6]$. The match probability is about 0.997. Same scores; the temperature did the work.

Same scores $[1.0, 0.6]$. Small τ makes the one negative brutal.



A batch of 2 still has only **one** negative. CLIP cannot emit "a cat sits on a mat" unless that string is already in your gallery. That generation job is BLIP.

6. Where students get stuck

- Asking CLIP to **write** a caption.
 - Fine-tuning a fused head and throwing away the text tower (you can no longer query with words).
 - Ignoring τ and wondering why a 2-D toy with $\tau = 1$ looks “softer” than the paper.
 - Using $[0.1, 0.9]$ and calling it unit length. Check $\|t\|_2 = 1$ before you trust the cosine.
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7. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#) (full paper video).

In class, pause on two towers, the InfoNCE matrix, zero-shot prompts, and the remark that prompt wording is engineering. Watch the rest after class.

8. Practice

1. CLIP’s text tower never sees the pixels. How can a typed query retrieve a photo?
2. Why does a batch of 8 give weaker negatives than a batch of 1024?
3. You need generated captions, not retrieval. Which Week 5 model is the closer ancestor?
4. For logits $[2, 0]$ (already divided by τ), compute the softmax probability on the positive. Repeat for $[2 / 0.07, 0]$. Which temperature makes the positive dominate?
5. Two gallery texts $t_{\text{cat}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $t_{\text{dog}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, query image $v = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ (already unit). Rank the texts by cosine. Who is retrieved at rank 1?